Year 10 Mathematics

Graphs 2

8.4 Perpendicular Lines

Perpendicular Lines

STARTER · 5 MIN

Date: _____ **Title:** Perpendicular Lines

LEARNING OBJECTIVES

- Know that perpendicular gradients multiply to −1
- Find the gradient of a perpendicular line
- Find the equation of a perpendicular line

STARTER *Answer in your book. No calculators.*

1. Write down the gradient of $y = 3x - 2$
2. Find the gradient through (1, 2) and (3, 8)
3. Work out $-1 \div \frac{2}{3}$
4. Solve $5 = 2 \times 4 + c$

Starter — Answers

STARTER

Tick or correct in green pen

1 $m = \mathbf{3}$

2 $m = \dfrac{8-2}{3-1} = \mathbf{3}$

3 $-1 \times \dfrac{3}{2} = \mathbf{-\dfrac{3}{2}}$

4 $5 = 8 + c, \quad c = \mathbf{-3}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I know that perpendicular gradients multiply to −1	☐
SILVER	I can find the gradient of a perpendicular line	☐
GOLD	I can find the equation of a perpendicular line	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Perpendicular Lines

INTRODUCTION · 3 MIN

Perpendicular lines meet at a right angle (90°).

Rule

The gradients multiply to give −1.

$m_1 \times m_2 = -1$

Negative reciprocal

Flip the fraction and change the sign.

$m = \frac{2}{3} \rightarrow -\frac{3}{2}$

Whole numbers

Write the number as a fraction first.

$m = 4 \rightarrow -\frac{1}{4}$

Equation

Find m, then substitute a point to find c.

$y = mx + c$

Examples 1 and 2 — Perpendicular Gradients

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $m_1 \times m_2 = -1$

Example 1

Find the gradient of a line perpendicular to

$y = 2x + 5$

Example 2

Find the gradient of a line perpendicular to a line with gradient $-\frac{3}{4}$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$m = 2$$

$$\text{Perpendicular} = -\frac{1}{2}$$

Tip: negative reciprocal: flip and change sign.

Example 2

$$m = -\frac{3}{4}$$

$$\text{Perpendicular} = \frac{4}{3}$$

Tip: negative × negative = positive.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the gradient of a line perpendicular to each line

Hint $m_1 \times m_2 = -1$

1. $y = 3x + 1$

2. $y = -5x + 2$

3. $y = \frac{1}{4}x - 3$

4. $y = -\frac{2}{5}x + 1$

5. $y = x$

★ A line with gradient -1

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $-\frac{1}{3}$

2 $\frac{1}{5}$

3 -4

4 $\frac{5}{2}$

5 -1

★ 1

Examples 3 and 4 — Are the Lines Perpendicular?

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $m_1 \times m_2 = -1$

Example 3

Are these lines perpendicular?

$y = 4x - 1$ and $y = -\frac{1}{4}x + 3$

Example 4

Are these lines perpendicular?

$y = 2x + 3$ and $2y = x - 4$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$4 \times -\frac{1}{4} = -1$$

Yes, they are perpendicular

Tip: multiply the gradients.

Example 4

$$2y = x - 4 \rightarrow y = \frac{1}{2}x - 2$$

$$2 \times \frac{1}{2} = 1$$

No, they are not perpendicular

Tip: rearrange to y = mx + c first.

Task 2

PRACTICE · 9 MIN

SILVER

State whether each pair of lines is perpendicular

Hint $m_1 \times m_2 = -1$

1. $y = 3x + 2$ and $y = -\frac{1}{3}x - 1$
2. $y = 5x$ and $y = -5x + 4$
3. $y = -2x + 1$ and $2y = x + 6$
4. $y = \frac{1}{2}x$ and $y = 2x - 3$
5. $x + y = 4$ and $y = x + 7$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $3 \times -\frac{1}{3} = -1$, Yes

2. $5 \times (-5) = -25$, No

3. $-2 \times \frac{1}{2} = -1$, Yes

4. $\frac{1}{2} \times 2 = 1$, No

5. $-1 \times 1 = -1$, Yes

Examples 5 and 6 — Equation of a Perpendicular Line

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $m_1 \times m_2 = -1$ $y = mx + c$

Example 5

(3 marks)

Find the equation of the line perpendicular to

$y = 2x + 1$ that passes through (4, 3).

Example 6

(3 marks)

Find the equation of the line perpendicular to

$y = -3x + 2$ that passes through (6, 1).

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$m = -\frac{1}{2}$$

$$3 = -\frac{1}{2} \times 4 + c$$

$$c = 5$$

$$\mathbf{y = -\frac{1}{2}x + 5}$$

Tip: find the perpendicular gradient first.

Example 6

$$m = \frac{1}{3}$$

$$1 = \frac{1}{3} \times 6 + c$$

$$c = -1$$

$$\mathbf{y = \frac{1}{3}x - 1}$$

Tip: substitute x and y from the point.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $m_1 \times m_2 = -1$ $y = mx + c$

1. Find the equation of the line perpendicular to $y = 4x - 2$ through $(8, 1)$. (3 marks)

2. Find the equation of the line perpendicular to $y = -2x + 5$ through $(2, 3)$. (3 marks)

3. Find the equation of the line perpendicular to $y = \frac{1}{3}x + 1$ through $(1, 4)$. (3 marks)

4. Find the equation of the line perpendicular to $2y = 6x + 1$ through $(3, 0)$. (3 marks)

5. Show that the lines $y = 5x - 2$ and $5y + x = 10$ are perpendicular. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $m = -\frac{1}{4}, \quad 1 = -2 + c, \quad y = \mathbf{-\frac{1}{4}x + 3}$

2. $m = \frac{1}{2}, \quad 3 = 1 + c, \quad y = \mathbf{\frac{1}{2}x + 2}$

3. $m = -3, \quad 4 = -3 + c, \quad y = \mathbf{-3x + 7}$

4. $m = -\frac{1}{3}, \quad 0 = -1 + c, \quad y = \mathbf{-\frac{1}{3}x + 1}$

5. $y = -\frac{1}{5}x + 2, \quad 5 \times -\frac{1}{5} = \mathbf{-1}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the gradient perpendicular to $y = 6x - 1$

SILVER Are $y = -4x + 2$ and $y = \frac{1}{4}x$ perpendicular?

GOLD Find the line perpendicular to $y = 2x$ through $(2, 5)$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $-\frac{1}{6}$

SILVER $-4 \times \frac{1}{4} = -1$, Yes

GOLD $m = -\frac{1}{2}$, $5 = -1 + c$, $y = -\frac{1}{2}x + 6$

Exam Practice — Perpendicular Gradients

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $m_1 \times m_2 = -1$

Question 1

(2 marks)

Write down the gradient of a line perpendicular to

(a) (1)

$y = 3x + 5$

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(b) (1)

$y = -\frac{1}{2}x + 2$

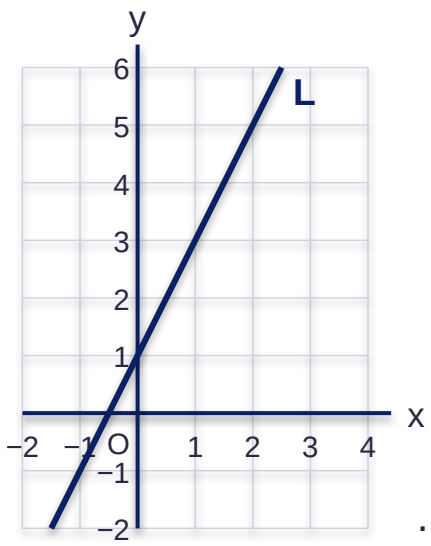
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Question 2

(2 marks)

Line L is drawn on the grid.

Find the gradient of a line perpendicular to L.



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Exam Practice — Are the Lines Perpendicular?

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $m_1 \times m_2 = -1$

Question 3

(3 marks)

Show that these lines are perpendicular.

$$y = 5x + 2$$

$$5y + x = 10$$

Question 4

(2 marks)

Here are the equations of three lines.

A: $y = 3x - 1$

B: $3y = x + 6$

C: $y = 4 - \frac{1}{3}x$

Which two lines are perpendicular?

Give a reason.

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Exam Practice — Equation of a Perpendicular Line

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $m_1 \times m_2 = -1$ $y = mx + c$

Question 5

(3 marks)

Find an equation of the line perpendicular to $y = 3x - 2$ that passes through the point (6, 1).

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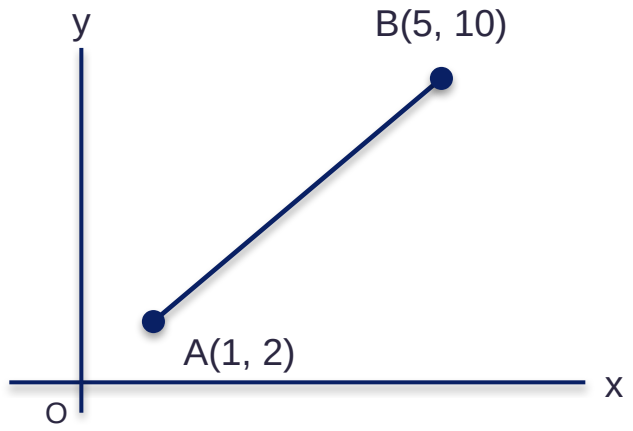
Question 6

(4 marks)

A is (1, 2) and B is (5, 10).

Find an equation of the perpendicular bisector of AB.

Diagram NOT accurately drawn



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Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $-\frac{1}{3}$ — B1

(b) 2 — B1

Tip: flip the fraction and change the sign.

Question 2

Gradient of L = 2 — M1

$-\frac{1}{2}$ — A1

Tip: $m_1 \times m_2 = -1$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$5y = -x + 10 \Rightarrow y = -\frac{1}{5}x + 2$ — M1

$5 \times -\frac{1}{5} = -1$ — M1

so perpendicular — A1

Tip: rearrange to y = mx + c first.

Question 4

B: $y = \frac{1}{3}x + 2$ — M1

A and C: $3 \times -\frac{1}{3} = -1$ — A1

Tip: test the product of each pair.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$m = -\frac{1}{3}$ — B1

$1 = -\frac{1}{3} \times 6 + c, \quad c = 3$ — M1

$\mathbf{y = -\frac{1}{3}x + 3}$ — A1

Tip: find the new gradient, then c.

Question 6

Mid-point (3, 6) — B1

Gradient AB = 2, perp. $= -\frac{1}{2}$ — M1

$6 = -\frac{1}{2} \times 3 + c$ — M1

$\mathbf{y = -\frac{1}{2}x + 7.5}$ — A1

Tip: bisector = through the mid-point, at 90°.